MINH-HOANG DANG



Seeking a position in Knowledge / Ontology Engineering, Linked Data & Enterprise Knowledge Graphs, Artificial Intelligence

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RESEARCH INTERESTS

Semantic Web, Knowledge / Ontology Engineering, Linked Data & Enterprise Knowledge Graphs, Artificial Intelligence, Music Information Retrieval

EDUCATION

PhD Computer Science, Semantic Web

Nantes University

Sept 2021 – November 2024

MSc Computer Science, Data Science

Nantes University

Sept 2019 – July 2021

CERTIFICATES

IBM Data Science
- Date Issued: September 07, 2024
- Credential: VZ6JZRK7L6XG

IBM AI Engineering
- Date Issued: July 6, 2024
- Credential: LV95FKMTPGHB

SKILLS

Large Language Models
- **Framework**: LangChain, llama-cpp-python, instructor
- **Platform**: Langfuse, Scorecard.io

Database
- MySQL, PostgresSQL, MongoDB, Spark.
- GraphDB, Neo4J, Virtuoso, Jena

Development
- Linux, Bash, Shell script, Docker, CI/CD

Semantic Web
- Protégé, Virtuoso, Jena, RDF4J
- Ontop (ODBA), Ontotext GraphDB

Cloud
- Google Cloud Platform, Azure (ML Studio)

Machine Learning
- SentenceTransformers, HuggingFace, Pytorch

LANGUAGES

English (Professional) | French (Bilingual) | Vietnamese (Native)

WORK EXPERIENCE

Natural Language Based Secure Federated Query Building

Inria, Wimmics team — **Postdoctoral Researcher**

June 2026 – June 2029 — Sophia Antipolis, France

- The grounding of LLM in the Knowledge Graph (GraphRAG) helps reduce hallucination in the generated answer.
- Develop a federated query builder and end-user interface, coupling generative models with linked data for natural language interaction.

Applied Research and Development Engineer

JusMundi — **Permanent**

November 2024 – May 2026 — Paris, France

- **The prior RAG system relied on hard-coded legal rules, reducing flexibility and increasing maintenance**. By building a knowledge-graph-based retrieval system using a domain ontology and RDB-to-RDF mappings, we replaced these rules, achieving comparable performance through graph-based retrieval and embeddings.
- **General-purpose embedding models struggled with legal jargon and limited domain data availability**. We synthetically generated queries from a large corpus of legal documents and used the resulting query-document pairs to fine-tune a long-context embedding model, achieving a 30%pt recall improvement for the Agentic RAG system.

Query the Web as a Knowledge Graph

Nantes University (LS2N) — **Doctoral Researcher**

September 2021 – November 2024 — Nantes, France

- **Can we trust LLMs to extract facts from the Web?** We proposed a new evaluation pipeline that carefully uses Schema.org ontology and documentation to verify LLMs' output in the task of extracting Knowledge Graphs from Web pages.
- **Can we still execute queries when the number of Knowledge Graphs increases?** First, we created a new benchmark called FedShop that tests the scalability of existing engines in terms of the number of sources. Second, we created a new SPARQL federation engine that finds better source assignment than existing engines.

E-mail personalisation through data

Visage.jobs — **Internship**

February 2021 – July 2021 — Nantes, France

- Using text similarity metrics such as Jaccard or Cosine, we learned that most users use template without personalization.
- Using text classification methods, we tagged sentences email to extract user writing patterns.
- Using Association Rule mining, we explain the factors that drive candidate engagement.

Attention-based abstractive summarisation on Yale courses

Nantes University — **Internship**

January 2020 – May 2020 — Nantes, France

- Proposed improvements for ROUGE, METEOR and human evaluation on educational content.

Data visualisation on blockchain network

Cogarius — **Internship**

April 2019 – June 2019 — Brussels, Belgium

- Visualised smart-contracts events on Ethereum; prototype web-application and graph database. Publication link.

Data visualisation on traffic network

Beuth University of Applied Science — **Internship**

April 2017 – June 2017 — Berlin, Germany

- Data visualisation on traffic network in Dresden (geo + time).

SCIENTIFIC PUBLICATIONS

LLM4Schema.org: Generating Schema.org Markups with Large Language Models

Semantic Web Journal

June 4th 2025

- **Generating a Schema.org markup is easy using LLMs, but Evaluating its quality is hard**. This paper introduces LLM4Schema.org, a novel approach to evaluate the performance of LLMs in generating Schema.org markups, mixing both deterministic and LLM-as-judge metric.
- Our findings indicate that LLMs should not be used out-of-the-box for generating Schema.org markups, as they often produce invalid, non-factual, or non-compliant markups.

FedUP: Querying large-scale federations of SPARQL endpoints

International World Wide Web Conference - Rank A*

May 17th 2024 — Singapore

- The FedShop benchmark shows that existing Federation Query Engines (FQEs) perform poorly when the number of endpoints exceeds few dozens. In Federated Querying settings, most of the query processing time is wasted evaluating on combinations of endpoints that return no results.
- We proposed FedUP, a new FQE that build Result-aware plans using quotient summary, resulting in 1-3 orders of magnitude improvement in execution time.

FedShop: A Benchmark for Testing the Scalability of SPARQL Federation Engines

International Semantic Web Conference 2023 - Rank A

November 6-10th 2023 — Athens, Greece

- Existing benchmarks test scalability of Federation Query Engines (FQEs) based on data volume but use a limited number of endpoints.
- Our new benchmark reveals scalability challenges for current FQEs, highlighting issues in source selection and query decomposition.

RAW-JENA: Approximate Query Processing for SPARQL Endpoints

International Semantic Web Conference 2023 - Rank A

November 6-10th 2023 — Athens, Greece

- Existing SPARQL endpoints enforces Fair-Use Policy, imposing a cap on the query processing time to user requests (60s on Wikidata). This limits the usability of these SPARQL endpoints in important use cases, e.g, exploratory queries, large-scale statistics, embeddings, etc.
- We present RAW-JENA: an extension of Apache Jena to support Sampling-based Approximate Query Processing for conjunctive SPARQL queries relying on random walks. RAW-JENA delivers partial random results and cardinality estimates in a pay-as-you-go fashion.

Schema.org: How is it used?

International Semantic Web Conference 2023 - Rank A

November 6-10th 2023 — Athens, Greece

- Due to the vast and diverse nature of the contributed annotations, it is not easy to understand the widespread use of Schema.org.
- Using Characteristic Sets, we established a comprehensive observatory for Schema.org annotations, visually presenting the most frequently used classes, commonly used combinations of properties per class, etc.

Join ordering of SPARQL property path queries

Extended Semantic Web Conference 2023 - Rank B

May 22nd 2023 — Hersonissos, Greece

- SPARQL Property Path Queries (SPPQ) provide a succinct way to write complex navigational queries over RDF knowledge graphs but their evaluation remains difficult due to transitive closures. We propose a novel query optimizer that uses an appropriate cost model and a sampling-based cardinality estimator to optimize join order of SPPQs with UNION and FILTER.
- The total execution time is 14x faster compared to BlazeGraph and 14% faster than Virtuoso on Wikidata Query Benchmark.

Improving Language Model Predictions via Prompts Enriched with Knowledge Graphs

Deep Learning For Knowledge Graph workshop at International Semantic Web Conference 2022 - Rank A

September 16th 2022 — Bertinoro, Italy

- Un-tuned Pre-trained language models are believed to perform poorly when it comes to Question-Aswering tasks.
- We argue that these un-tuned models already know a lot. By improving the prompt, we can improve the performance.
- We use Wikidata Knowledge Graph to augment the prompt and improve the BERT/RoBERTa base models in Movie Question-Answering task.